A novel reliability aware patch selection based few shot learning for target recognition and open-set identification in ill posed SAR images

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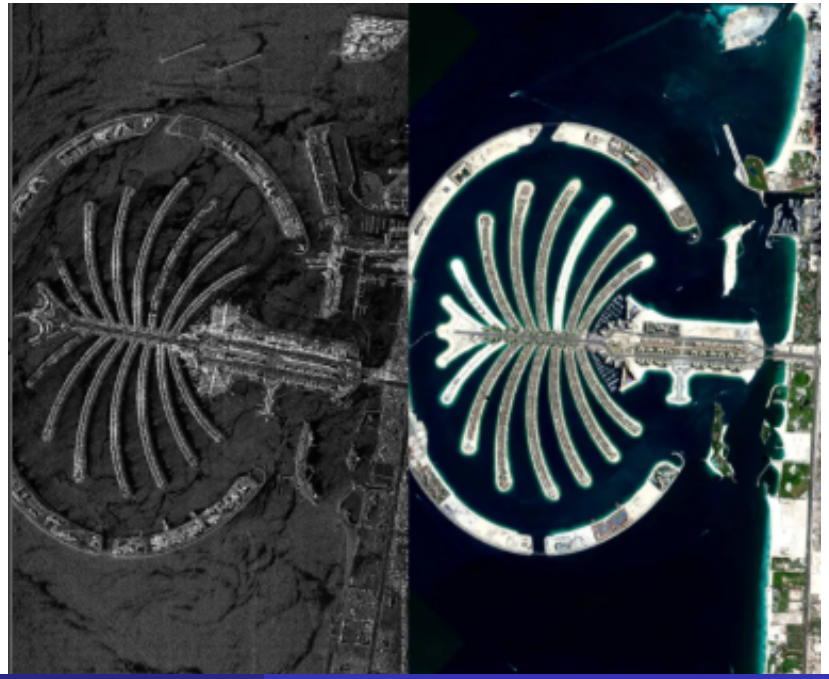
Introduction: SAR and Classification Challenges

- **Synthetic Aperture Radar (SAR):** Active remote sensing using microwave pulses for high-resolution imaging. Operates day/night and in all-weather conditions.
- **The Challenge:** SAR ATR is difficult due to:
  - High speckle noise from microwaves
  - Complex backscattering (surface, material, moisture, angle)
  - Lack of color/texture cues vs optical images [8, 9]



Why SAR Images Are Important

SAR (Synthetic Aperture Radar)	**Optical Images**
Operates day and night (active microwave sensing).	Dependent on sunlight, cannot capture images at night.
Penetrates clouds, fog, and rain (all-weather imaging).	Weather-dependent, obstructed by clouds and atmospheric conditions.
Captures structural, moisture, and roughness information.	Provides natural color, texture, and visually intuitive details.
Contains speckle noise and geometric distortions (harder to interpret).	High visual clarity but limited to surface reflectance only.

Key Advantage: SAR provides critical all-weather, all-time surveillance capability essential for defense and environmental monitoring.

Literature Review: Key Approaches (1/2)

Sl.No.	Title	Venue / Year	Key Points	Drawback
1	SAR image classification based on multi-feature fusion decision CNN [12]	IET Image Process. (2022)	• Proposes a multi-feature fusion decision CNN. • Uses multiple handcrafted and deep features.	High model complexity, risk of overfitting.
2	Using VGG16 for Military Target Classification [10]	Proc. IEEE Conf. (2021)	• Applies pre-trained VGG16 to SAR target recognition. • Achieves strong performance on MSTAR.	Requires large memory, computationally heavy.
3	A Novel Lightweight CNN for Constrained IoT Devices [11]	IEEE Access (2024)	• Extremely small model suitable for IoT hardware. • High accuracy with parameter efficiency.	Underperforms compared to deeper models.

Literature Review: Key Approaches (2/2)

Sl.No.	Title	Venue / Year	Key Points	Drawback
4	MHD-Protonet: Margin-Aware Hard Example Mining [13]	Algorithms (2025)	• Introduces margin-aware hard example mining. • Employs dual-loss optimization.	Complex training pipeline, performance sensitive.
5	Few-Shot SAR ATR via Multilevel Contrastive Learning [1]	IEEE J-STARS (2025)	• Multilevel Contrastive Learning (MCL). • Dependence Matrix (DMM).	Computational overhead. Uses single feature for opposing jobs.

Research Gaps & Problem Statement

Research Gaps:

- **Threshold Setting:** Hard to balance rejection vs misclassification
- **Lack of Benchmarks:** No standard open-set few-shot SAR ATR setup
- **Poor Uncertainty:** Overconfident predictions blur known vs unknown
- **Missing Local Patterns:** Ignores fine-grained SAR structures

Problem Statement:

- **Given:**
  - A small, labeled SAR dataset: $D_L = \{(x_i, y_i)\}_{i=1}^{n_L}$
  - A large, unlabeled SAR dataset: $D_U = \{x_j\}_{j=1}^{n_U}$

  where the number of unlabeled samples is much greater than the labeled ones ($n_U \gg n_L$).

  Objective:

  To learn a classifier, $f : X \to \{1, ..., K\}$, that can accurately predict labels y_j for the unlabeled image x_j that handlles the scarcity, imbalancing and open-set recognition too.

Objectives

1. Develop novel few-shot learning framework for SAR targets with limited labeled samples
2. Design strategies mitigating class imbalance using local patch feature identification
3. Propose unified pipeline handling data scarcity and zero-shot novelty
4. Reduce misclassification among similar targets via patch-level analysis
5. Enable recognition of previously unseen target classes through generalizable features

Proposed Methodology

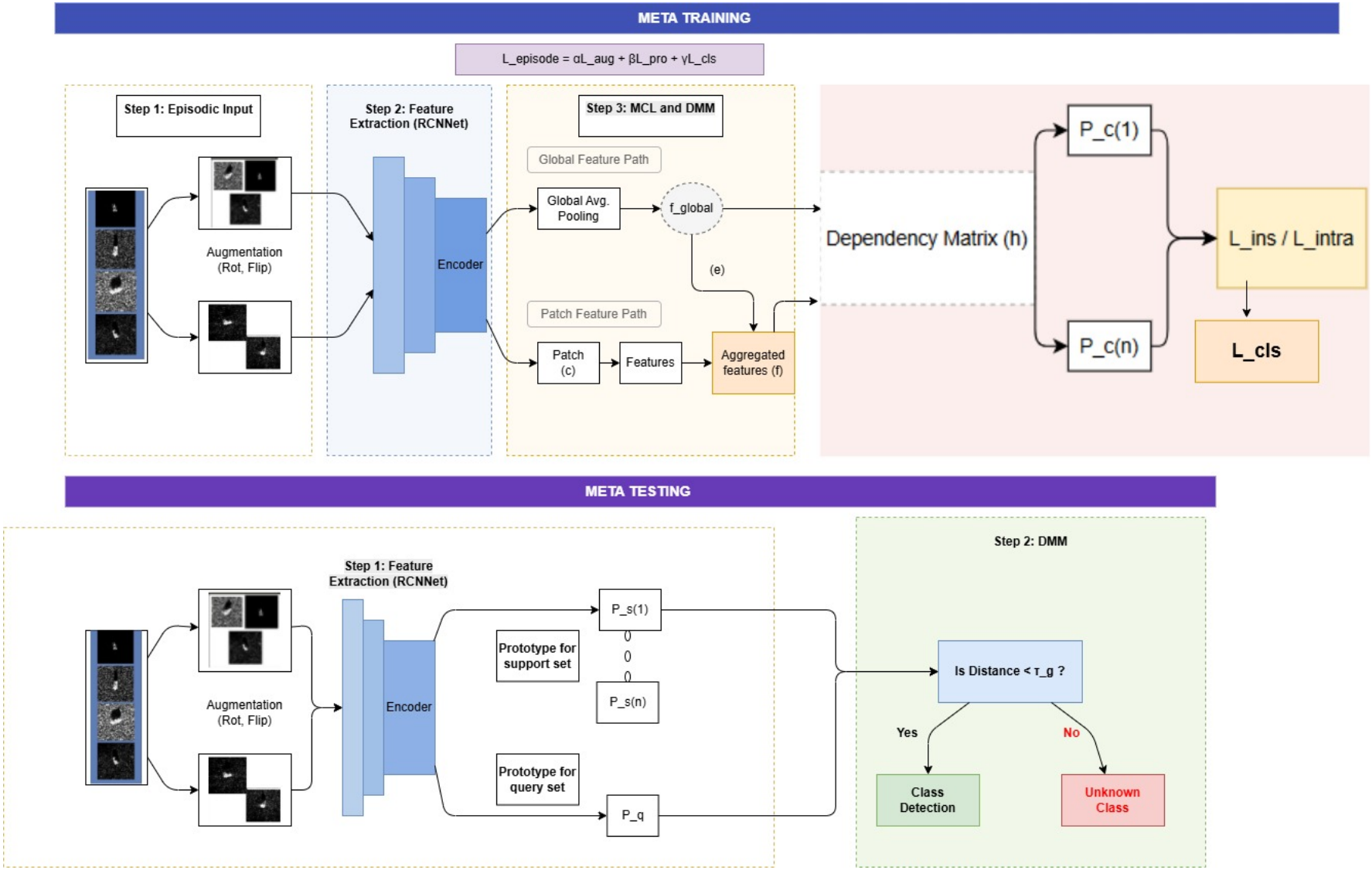


Figure: Our Proposed Workflow: (1) (Global+Patch) Feature Extraction → (2) MCL Training → (3) DMM Classification.

Core Methodology and DMM Overview

Supporting Components:

- **Few-Shot Learning:** N-way K-shot with support (train) and query (test)
- **Patch Selector:** 3×3 entropy filtering → removes noisy regions
- **Discrimination:** Cluster (class-level) + Individual (sample-level)

DMM: Dependence Matrix-Based Measurement (Key Contribution)

- Models **channel-wise correlations** within feature maps
- Builds matrices for **query** and each **class**
- Uses **Euclidean norm distance** for similarity comparison
- Captures **relational structure** → robust to noise & variation

Open-Set Recognition Framework

Phase 1: Embedding Space Construction

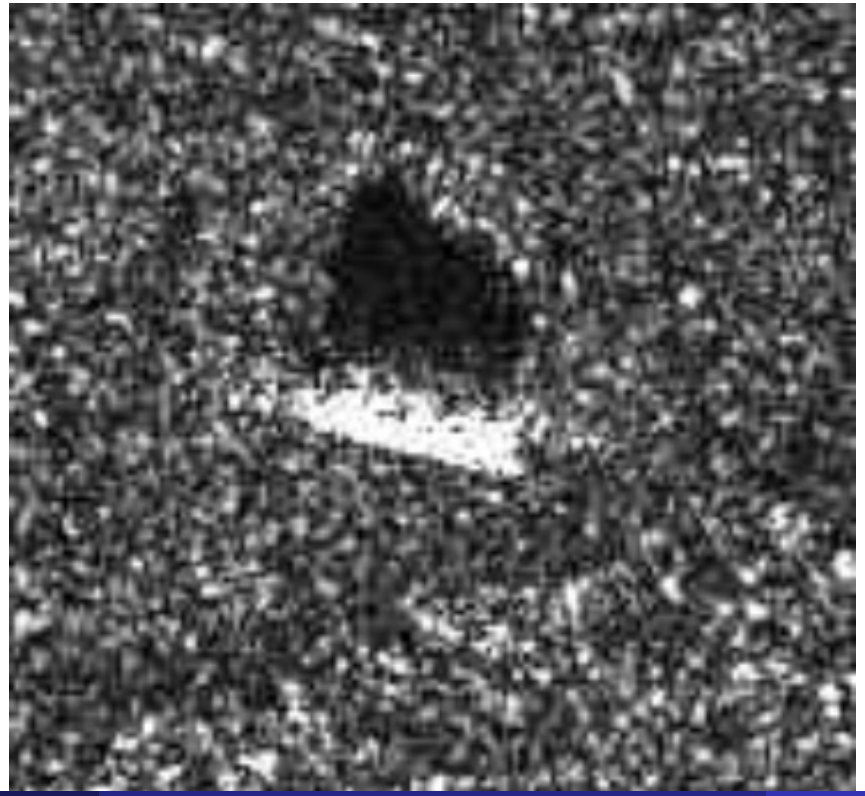
- Learn discriminative metric space from 7 base SAR vehicle classes
- Three-Block Encoder: RCNNet backbone → MCL contrastive head → DMM metric module
- MCL Loss pulls same-class together, pushes different-class apart
- Compute class prototypes as mean of training embeddings

Phase 2: Zero-Shot Inference

- **Novel Classes:** Seen (2S1, BRDM2 with support samples) + Unseen (ZSU_23_4 without samples)
- **Episodic Setup:** Support set (K shots per seen class) + Mixed query set
- **Frozen Encoder:** No retraining; compute episodic prototypes from support
- **Adaptive Threshold:** Calibrated from support cluster tightness (mean + ×std of intra-class distances, 1.99)
- **Decision Rule:** Compute distance to nearest prototype; if > threshold → reject as unseen, else assign to nearest class

Implementation: MSTAR Dataset

- **Dataset:** MSTAR (Moving and Stationary Target Acquisition and Recognition)
- Standard public benchmark for SAR ATR
- Contains 10 classes of military vehicles at different aspect and depression angles



Ablation Study & Benchmark Comparison

Ablation Study under Few-Shot Settings

Absent Module	**3-Way 1-Shot (%)**		**3-Way 5-Shot (%)**	
	MSTAR	SAMPLE	MSTAR	SAMPLE
Patchification	81.53	80.42	86.28	84.50
Entropy Filtering	84.39	82.56	88.25	86.54
Cluster Consistency Loss	86.61	84.59	90.16	88.67
Dependency Matrix	87.25	85.32	91.08	89.92
Proposed (Full)	**91.12**	**90.58**	**95.92**	**94.26**

Benchmark Comparison on Imbalanced Few-Shot SAR Classification

Method	**MSTAR (%)**				**SAMPLE (%)**			
	K=1	K=3	K=5	K=7	K=1	K=3	K=5	K=7
ConvT [6]	77.76	79.78	81.05	82.43	76.12	80.82	82.15	82.56
Meta-FSCIL [7]	81.09	83.34	83.82	84.54	78.05	81.15	82.83	86.79
Position-aware [5]	81.76	86.15	87.56	89.87	79.70	83.07	85.78	89.50
MHD-ProtoNet [13]	88.67	90.56	91.87	92.06	88.09	88.53	90.59	90.72
MCL-DMM [1]	90.79	92.17	93.05	93.97	89.73	91.55	91.82	92.75
Proposed	**91.12**	**94.84**	**95.92**	**96.24**	**90.58**	**92.95**	**94.26**	**94.89**

Open-Set Performance & Conclusion

Open-Set Recognition Performance

Method	**Known (%)**		**Unknown (%)**	
	MSTAR	SAMPLE	MSTAR	SAMPLE
Incremental [2]	90.32	88.76	73.71	69.93
Threshold-free [3]	89.10	87.45	70.60	68.97
Capsule + KLD [4]	88.87	89.12	71.89	69.28
Proposed	**91.42**	**90.01**	**79.81**	**74.56**

Conclusion:

- Novel Global+Patch fusion: +10 percentage point gain in 1-shot scenario
- **Successfully got it published at IEEE WACV, 2026:** Paper Link

- **Future Work:** Test on more complex SAR datasets and optimize patch-selection mechanism

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